

DEVELOPER BUILD GUIDE

SMAART Career Intelligence Platform

Complete Solution Design for Role Data Generation, Student Matching,
Role Changes, Learning Paths, and Skills Passport Generation
Step-by-Step Instructions with All Exceptions, Testing Procedures, and Scale Design

Target Scale: 50,000 Students | 25 Colleges | 250 Roles

Written in Plain English | Every Step Explained | All Edge Cases Covered

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Part 1: System Overview and Architecture

This document tells you exactly how to build the core data and matching engine for the SMAART Career Intelligence Platform. It covers everything from generating the role data to matching students with roles to handling role changes to generating Skills Passports. Follow it step by step and you will have a working system that serves 50,000 students.

1.1 What the System Does (The Big Picture)

The platform helps Indian college students figure out which careers are right for them and what they need to learn. It does this through five things: it stores detailed profiles for 250 career roles (what skills each role needs, which AI tools are used, how exposed the role is to AI, salary ranges, and a plain-English description). It matches each student to 3 recommended roles based on their degree, specialisation, and career interests. It tracks which skills each student has developed and shows them a personalised learning path for their chosen role. It lets students change roles freely and instantly recalculates everything. And it generates a verified Skills Passport when a student is ready.

1.2 The Critical Design Principle

✓ AI (Claude API) is used **ONLY** during the **BUILD** phase to generate data. At **RUNTIME** when students use the platform, everything is database queries. Zero AI calls for browsing roles, changing roles, viewing learning paths, or checking skill coverage. The only runtime AI calls are for quotient assessment scoring and Skills Passport narrative generation, which are rare and cached.

This means the platform costs almost nothing to run at any scale. 50,000 students or 500,000 students — the database queries cost the same.

1.3 What You Are Building

Component	What It Does	Uses AI?	Build Time
Role Data (250 profiles)	Stores all information about each career role — skills, tools, salary, AI exposure, narrative	Yes — Claude API during generation only. Once generated and verified, stored as static data	7-8 days
Matching Engine	Suggests 3 roles (Primary, Secondary, Alternative) based on student's degree, specialisation, and interest	Yes — Claude API to generate the matching table once. Then pure database queries at runtime	3-4 days
Role Change System	Lets students switch between roles freely, archives old roles, recalculates learning paths instantly	No — pure database queries	2-3 days
Learning Path Engine	Calculates prioritised list of skills and courses for each student's current role	No — arithmetic formula applied to database data	2-3 days
Skills Passport	Creates a verified capability document for students who meet the	Yes — one Claude API call per passport,	2-3 days

Generator	eligibility threshold	cached	
Engagement Score Calculator	Calculates a weekly score showing how actively each student uses the platform	No — formula applied to database data	1-2 days

Total build time: approximately 20-24 working days for one developer. These can be built in parallel if you have two developers — one on data generation and matching, one on role changes and learning paths.

1.4 The Database Tables Involved

You already have the 8-table schema from the main system design document. This guide adds 3 new tables and modifies 2 existing ones. Here is the complete list of tables this guide touches:

Table	Purpose	New or Existing
roles	Stores all 250 role profiles — name, job family, skills count, AI exposure data, salary, narrative	Existing — add new columns
role_skills	Maps which skills belong to which role, with importance level	Existing
skills	Master list of all skills (technical + AI tools + human)	Existing
courses	Free courses mapped to skills	Existing
students	Student profiles including current role, degree, specialisation	Existing — add new columns
student_skill_completions	Tracks which skills each student has completed	Existing
degree_role_recommendations	Pre-built matching table: degree + specialisation + interest → 3 recommended roles	NEW
degree_interest_areas	Dropdown options for each degree-specialisation combination	NEW
degree_role_zones	Green/Amber/Red zone classification for each degree-role pair	Existing from career change design
role_archive	Stores snapshot when student switches away from a role	NEW

Part 2: Generating Role Data Using Claude API

This is the first thing you build. Before students can browse roles, before the matching engine works, before learning paths exist — you need the role data. This section tells you exactly how to generate accurate, verified data for 250 roles using the Claude API.

2.1 Why Claude and Not ChatGPT Mini

Your previous attempt used GPT-4o-mini, which produced three types of errors: it invented AI tool names that do not exist (hallucination), it listed generic skills that apply to every role instead of skills specific to each role, and it wrote vague narratives with no real information. GPT-4o-mini is too small and too unreliable for structured data generation. Claude Sonnet is the right model for this task.

2.2 Cost Calculation (Read This First)

The total cost to generate ALL 250 role profiles is approximately \$9 (₹750). Not per role — total. Here is the calculation:

Item	Tokens	Rate	Cost
Step 1 input (prompt) x 250 roles	200,000 tokens	\$3 per million tokens	\$0.60
Step 1 output (JSON) x 250 roles	375,000 tokens	\$15 per million tokens	\$5.63
Step 2 input (prompt + JSON) x 250 roles	250,000 tokens	\$3 per million tokens	\$0.75
Step 2 output (narrative) x 250 roles	125,000 tokens	\$15 per million tokens	\$1.88
TOTAL	950,000 tokens		\$8.86 (approximately ₹750)

Even if you run it 3 times to compare outputs, the total is under ₹2,500. This is less than the cost of one team lunch. Do not try to save money by using a cheaper, less accurate model.

2.3 API Setup

Step 1: Go to console.anthropic.com and create an account.

Step 2: Navigate to API Keys and create a new key. Copy it and store it securely. You will use this key in all API calls.

Step 3: Go to Billing and add \$10 of credits (the minimum top-up). This \$10 will last you through multiple rounds of generation and testing.

Step 4: Test the connection with a simple API call to confirm your key works:

```
curl https://api.anthropic.com/v1/messages \
  -H "Content-Type: application/json" \
  -H "x-api-key: YOUR_API_KEY" \
  -H "anthropic-version: 2023-06-01" \
```

```
-d '{"model":"claude-sonnet-4-20250514","max_tokens":100,
  "messages":[{"role":"user","content":"Say hello"}]}'
```

If you get a response with "Hello", your API is working. If you get an error, check your API key and billing status.

2.4 The Two-Step Generation Process

Do NOT ask the AI to generate everything in one prompt. That is what causes quality to drop. Instead, use two separate steps: Step 1 generates the structured data (skills, tools, salary, AI exposure) as JSON. Step 2 generates the narrative text using the Step 1 JSON as input. This separation ensures the AI focuses on accuracy in Step 1 and readability in Step 2.

STEP 1 PROMPT — Structured Data (Copy This Exactly)

Replace only the parts in square brackets. Everything else stays exactly as written.

You are a career intelligence analyst specialising in the Indian job market. Generate accurate, specific data for the following role. Do NOT invent or hallucinate any information. If you are uncertain about something, say "needs verification" instead of guessing.

ROLE: [Financial Analyst]

EXPERIENCE LEVEL: 0-3 years (entry level, fresh graduate)

COUNTRY: India

Generate the following in valid JSON format:

1. "technical_skills": List exactly 8-10 technical skills that are SPECIFIC to this role in India. Do NOT list generic skills like "communication" or "teamwork". Only TECHNICAL skills. For each skill include:
 - "name": the skill name
 - "importance": "High" or "Medium" or "Low"
 - "description": one sentence explaining this skill
2. "ai_tools": List exactly 4-6 AI tools that professionals in this role in India are ACTUALLY using in 2025-2026. Do NOT list tools that are theoretical. Only tools that appear in real job postings. For each tool include:
 - "name": the exact tool name
 - "category": "AI-Powered" or "Pure AI"
 - "usage": one sentence on how this tool is used in this role
 - "adoption": "Widely Used" or "Growing" or "Emerging"
3. "human_skills_priority": List the top 5 human/soft skills ranked from most to least important. Choose ONLY from this list: Analytical Thinking, Resilience and Adaptability, Leadership, Creative Thinking, Motivation and Self-Awareness, Empathy and Active Listening, Curiosity and Lifelong Learning, Execution and Reliability, Collaboration, Stakeholder Communication.

4. "salary_range":
 - "low": annual salary in INR for entry level in Tier 2 cities
 - "high": annual INR for 2-3 years experience in Tier 1 cities
5. "ai_exposure":
 - "percentage": estimated % of daily tasks assisted by AI
 - "level": "Low" (0-24%) or "Moderate" (25-50%) or "High" (51%+)
 - "detail": one sentence on which parts AI handles
 - "human_value": one sentence on which parts CANNOT be done by AI
6. "emerging_role_connection": If this role connects to emerging roles, name 1-2 and explain in one sentence each. If no strong connection, say "none".

Output ONLY the JSON. No explanations before or after.

STEP 2 PROMPT — Narrative Text (Copy This Exactly)

You are writing for Indian college students aged 18-24 who are exploring career options. Write in simple, clear English.

Based on this role data:

[PASTE THE JSON OUTPUT FROM STEP 1 HERE]

Write THREE short paragraphs for the role of [Financial Analyst]:

PARAGRAPH 1 - "What This Role Actually Does" (4-5 sentences):
Describe the day-to-day work of a junior [Financial Analyst] in an Indian company. Be specific and practical. Mention actual tasks, who they report to, and what a typical week looks like.

PARAGRAPH 2 - "How AI Is Changing This Role" (3-4 sentences):
Based on the ai_exposure data, explain how AI is affecting this role. What tasks does AI help with? What still needs a human?

PARAGRAPH 3 - "Who Should Consider This Role" (3-4 sentences):
Based on the skills data, describe the type of person who would succeed in this role. What mindset do they need? Which degree backgrounds typically lead here?

Keep each paragraph under 80 words. Total under 250 words.
No bullet points. No headers. Just three flowing paragraphs.

2.5 Reducing AI Tool Hallucination

⚠ AI tool names are the area where models hallucinate most. A model might invent 'FinanceAI Pro' that does not exist. Use the approved tool list below to constrain the output.

Before running generation, create a master list of verified AI tools. Add this instruction to the Step 1 prompt, immediately before 'Output ONLY the JSON':

For `ai_tools`, STRONGLY PREFER tools from this approved list:
 Excel with Copilot, GitHub Copilot, ChatGPT, Claude, Gemini,
 Power BI, Tableau, Google Analytics, Canva AI, Jasper AI,
 HubSpot AI, Salesforce Einstein, SAP AI, Zoho AI, Freshworks
 AI, Tally (with AI features), QuickBooks AI, Grammarly AI,
 Notion AI, Figma AI, Adobe Firefly, Midjourney, DALL-E,
 Google Workspace AI, Microsoft 365 Copilot, Slack AI,
 Jira AI, Confluence AI, Trello AI, Loom AI, Otter.ai,
 Beautiful.ai, Gamma.app, Copy.ai, Writesonic, SurferSEO,
 SEMrush AI, Ahrefs AI, Mailchimp AI, Hootsuite AI.

If you believe an important tool is missing from this list,
 you may include it but add `"needs_verification": true`.

2.6 Running the Generation (Two Options)

Option A: Manual via Claude.ai (Simplest, 1-2 Days)

Step 1: Open claude.ai and log in with a Pro subscription (\$20/month).

Step 2: Paste the Step 1 prompt with the first role name. Wait for the JSON response. Copy the JSON into a text file or spreadsheet.

Step 3: Paste the Step 2 prompt with the Step 1 JSON included. Wait for the narrative. Copy it.

Step 4: Repeat for each role. At approximately 3 minutes per role times 2 steps, 250 roles takes about 25 hours of work — approximately 3 working days.

This is boring but reliable. No coding needed. It works.

Option B: API Script (Faster, Requires Coding)

Write a script in your preferred language (Python, Node.js, etc.) that automates the process. Here is the logic in plain English:

```
Read "roles.csv" (one column: role_name)

For each role_name:
    prompt1 = Step 1 prompt template with role_name inserted
    response1 = call Claude API (model: claude-sonnet-4-20250514,
                                max_tokens: 2000, messages: [{role: "user",
                                                                content: prompt1}])
    step1_json = response1.content[0].text

    Wait 1 second (avoid rate limits)

    prompt2 = Step 2 prompt template with role_name and
              step1_json inserted
    response2 = call Claude API (model: claude-sonnet-4-20250514,
                                max_tokens: 1000, messages: [{role: "user",
                                                                content: prompt2}])
    step2_narrative = response2.content[0].text
```

```
Append to "output.csv": role_name, step1_json, step2_narrative
Wait 1 second
```

```
Print "Done. Generated data for X roles."
```

The script runs through 250 roles in approximately 40 minutes. Total API cost: approximately \$9.

2.7 Verification Process (Do NOT Skip This)

Even with Claude Sonnet producing much better output than GPT-4o-mini, you must verify the data before loading it into the platform. Here is the verification process in priority order:

Verification 1: AI Tools (5 hours for 250 roles)

For each role's AI tools list (4-6 tools), do a quick Google search for the tool name. If the tool exists and is used in Indian workplaces, keep it. If you cannot find evidence of it existing, remove it. Any tool marked 'needs_verification: true' MUST be searched. Time: approximately 30 seconds per tool.

Verification 2: Salary Ranges (6 hours for 250 roles)

⚠ AI-generated salary ranges are frequently wrong. Do NOT trust them. Verify every single one.

For each role, search AmbitionBox (ambitionbox.com/salaries) for the role name with 0-3 years experience. Update the salary_range_low and salary_range_high with the actual data. Time: approximately 2 minutes per role.

Verification 3: Technical Skills (4 hours for 250 roles)

Quickly scan each role's skill list. Check: are these skills SPECIFIC to this role or could they apply to any role? If 'Microsoft Office' appears in every role, it is too generic — remove it and replace with something specific. If you see 'Financial Statement Analysis using Ind AS' for Financial Analyst, that is correctly specific. Time: approximately 1 minute per role.

Verification 4: AI Exposure (3 hours for 250 roles)

For roles that have a matching O*NET occupation code in the Anthropic Economic Index dataset (downloaded from Hugging Face), use the Anthropic number instead of the model's estimate. For roles without a match, the model's estimate is a reasonable starting point — label it as 'estimated' in the database.

Total verification time: approximately 18 hours, or 2.5 working days. Combined with generation time, the complete process takes 5-6 working days for 250 roles.

2.8 Loading Data Into the Database

After verification, load the data into the database. Here is the exact process for each role:

Step 1: Insert or update a row in the roles table with: role_name, job_family, ai_exposure_pct, ai_exposure_level, ai_exposure_detail, human_value_tasks, salary_range_low, salary_range_high, narrative_text, emerging_role_connection.

Step 2: For each technical skill in the role's JSON: check if the skill already exists in the skills table (it may be shared across roles). If not, insert it. Then insert a row in role_skills mapping the skill to the role with the importance level.

Step 3: For each AI tool in the role's JSON: insert it as a skill with skill_category = 'AI-Tool'. Map it to the role in role_skills.

Step 4: For each human skill in the priority list: these skills already exist in the skills table (they are the same across all roles — Analytical Thinking, etc.). Insert a row in role_skills mapping the skill to the role with the priority rank.

Step 5: For each technical skill, find a free course that teaches it and insert a row in the courses table (if the course does not already exist). Map the course to the skill.

2.9 Testing the Role Data

After loading, run these tests to confirm the data is correct:

Test	Expected Result	How to Check
Count roles	250 rows in the roles table	SELECT COUNT(*) FROM roles WHERE is_active = TRUE
Skills per role	Every role has 12-20 skills (8-10 technical + 4-6 AI tools + 5 human)	SELECT role_id, COUNT(*) FROM role_skills GROUP BY role_id — check no role has fewer than 10 or more than 25
No orphan skills	Every skill in role_skills exists in the skills table	SELECT * FROM role_skills WHERE skill_id NOT IN (SELECT skill_id FROM skills) — should return 0 rows
AI exposure consistency	Every role with ai_exposure_pct between 0-24 has level 'Low', 25-50 has 'Moderate', 51+ has 'High'	SELECT * FROM roles WHERE (ai_exposure_pct <= 24 AND ai_exposure_level != 'Low') OR ... — should return 0 rows
Salary sanity	No role has salary_range_low > salary_range_high. No role has salary below 100000 or above 3000000 for entry level	Query for violations — should return 0 rows
Narrative exists	Every role has non-empty narrative_text	SELECT * FROM roles WHERE narrative_text IS NULL OR narrative_text = "" — should return 0 rows
Course coverage	At least 80% of technical skills have at least one mapped course	Count skills with no course mapping — should be under 20% of total skills

Part 3: The Student-to-Role Matching Engine

When a student signs up and enters their degree, specialisation, and career interests, the platform instantly suggests 3 roles: Primary (strongest match), Secondary (good alternative), and Alternative (different direction using transferable skills). This section explains exactly how to build this.

3.1 How It Works (No AI at Runtime)

✓ The matching engine is a **PRE-BUILT TABLE**, not a real-time AI call. You generate the table once using Claude during the build phase. At runtime, matching is a single database query that returns 3 rows. Instant. Free. Works for 50,000 simultaneous users.

3.2 The Three New Tables

Table: degree_interest_areas

Column	Type	Description
id	Auto-increment integer	Unique ID
degree_type	Text	e.g., 'B.Com', 'B.Tech', 'BA', 'MBA'
specialisation	Text	e.g., 'Finance', 'Computer Science', 'English'
interest_area_name	Text	e.g., 'Banking and Financial Services'
display_order	Integer	Order to show in dropdown (1, 2, 3...)

This table powers the dropdown that the student sees during onboarding. Each degree-specialisation combination has 5-6 interest areas plus a 'General / Exploring Options' option. Total rows: approximately 400-600 across all degree-specialisation combinations.

Table: degree_role_recommendations

Column	Type	Description
id	Auto-increment integer	Unique ID
degree_type	Text	e.g., 'B.Com'
specialisation	Text	e.g., 'Finance'
interest_area	Text	Must match a value in degree_interest_areas
role_id	Foreign key to roles	The recommended role
recommendation_type	Text	'Primary' or 'Secondary' or 'Alternative'
match_strength	Integer 1-100	How strong the match is
reason	Text	One sentence explaining the match

This is the core matching table. Each degree-specialisation-interest combination has exactly 3 rows (Primary, Secondary, Alternative). Total rows: approximately 1,200-1,800.

Table: role_archive

Column	Type	Description
id	Auto-increment integer	Unique ID
student_id	Foreign key to students	Which student
role_id	Foreign key to roles	The role they switched away from
skills_covered_at_archive	Integer	How many skills they had covered when they left
total_skills_at_archive	Integer	How many total skills the role had when they left
days_on_role	Integer	How many days they spent on this role
archived_date	Date	When they switched away

3.3 Generating the Matching Table

Use Claude to generate the matching data for each degree-specialisation combination. Here is the exact prompt:

```
You are a career counsellor for Indian college students.
```

```
DEGREE: [B.Com]
```

```
SPECIALISATION: [Finance]
```

```
EXPERIENCE LEVEL: Fresh graduate (0-1 years)
```

```
For each interest area below, recommend exactly 3 roles from
the approved role list. One Primary (strongest match), one
Secondary (good alternative), one Alternative (different
direction using transferable skills).
```

```
Interest Area 1: "Banking and Financial Services"
```

```
Interest Area 2: "Corporate Finance and Accounting"
```

```
Interest Area 3: "Data and Analytics"
```

```
Interest Area 4: "Sales and Marketing"
```

```
Interest Area 5: "General / Exploring Options"
```

```
For each recommendation provide:
```

- interest_area: the interest area name
- role_name: exact role name from the approved list
- recommendation_type: Primary / Secondary / Alternative
- match_strength: 1-100
- reason: one sentence explaining the match

```
APPROVED ROLE LIST (choose ONLY from these):
```

```
[PASTE YOUR COMPLETE LIST OF 250 ROLE NAMES HERE]
```

```
Output as JSON array. No explanations.
```

Run this for each of your 80-120 degree-specialisation combinations. Total API cost: approximately \$3-5. Total time: 20-30 minutes via API script.

3.4 How Matching Works at Runtime

When a student onboards, the system runs these steps:

Step 1: Student selects their degree (dropdown) and specialisation (dropdown). The platform queries degree_interest_areas: 'Get all interest areas where degree_type = B.Com AND specialisation = Finance, ordered by display_order.' Returns 5-6 options.

Step 2: Student selects an interest area from the dropdown (e.g., 'Banking and Financial Services').

Step 3: The platform queries degree_role_recommendations: 'Get all rows where degree_type = B.Com AND specialisation = Finance AND interest_area = Banking and Financial Services.' Returns exactly 3 rows: one Primary, one Secondary, one Alternative.

Step 4: The platform displays: 'Based on your B.Com Finance degree and interest in Banking, we recommend: Primary — Financial Analyst (match: 92%), Secondary — Credit Analyst (match: 78%), Alternative — Business Analyst (match: 65%).' Each recommendation shows the reason sentence.

Step 5: Student clicks 'Accept' on one of the 3 recommendations, OR clicks 'Browse All Roles' to choose something else from the full 250-role catalogue.

Step 6: The selected role becomes the student's primary_role_id in the students table. The role_start_date is set to today.

✓ **Total runtime queries: 2 (interest areas + recommendations). Total API calls: 0. Response time: under 200 milliseconds. Works for 50,000 simultaneous users.**

3.5 Students Who Already Have Technical Skills

During onboarding, students also list their existing technical skills. After the initial 3-role suggestion, the platform runs an additional check:

```
SELECT r.role_name, COUNT(rs.skill_id) as matching_skills
FROM roles r
JOIN role_skills rs ON r.role_id = rs.role_id
JOIN student_onboarding_skills sos ON rs.skill_id = sos.skill_id
WHERE sos.student_id = [this student]
      AND r.role_id NOT IN (already suggested roles)
GROUP BY r.role_id
HAVING COUNT(rs.skill_id) >= 2
ORDER BY matching_skills DESC
LIMIT 3
```

This finds roles where the student already has 2 or more matching skills that were NOT in the original 3 suggestions. These appear as a 'You Might Also Consider' section below the main recommendations. This is optional — if no matches are found, the section is not shown.

3.6 The 'General / Exploring Options' Fallback

Many students do not know what they want. When they select 'General / Exploring Options', the matching table returns the 3 most popular roles for their degree-specialisation based on job market demand. These are the roles with the most job postings on Naukri for graduates of their degree type.

For example, a B.Com Finance student who selects 'General' might get: Primary — Accountant (the most common destination), Secondary — Financial Analyst (the second most common), Alternative — Business Analyst (a cross-functional option). These defaults are set in the matching table during generation and can be updated based on placement data from your colleges.

3.7 Testing the Matching Engine

Test	Expected Result	How to Check
Every degree-specialisation has interest areas	No degree-specialisation in your college portfolio has zero interest areas	Query degree_interest_areas grouped by degree+specialisation — every combination should have 5-7 rows
Every interest area has 3 recommendations	Every combination of degree+specialisation+interest has exactly 3 rows in degree_role_recommendations	GROUP BY degree_type, specialisation, interest_area, COUNT(*) — every group should have exactly 3
One Primary, one Secondary, one Alternative	Within each group of 3, exactly one of each type exists	Query for groups where the count of each recommendation_type is not exactly 1 — should return 0
All recommended roles exist	Every role_id in degree_role_recommendations exists in the roles table	JOIN check — 0 orphans
Match strength ranges	All match_strength values are between 1 and 100. Primary is always highest, Alternative is always lowest	Query for violations
'General' exists for every degree-specialisation	Every degree-specialisation has a 'General / Exploring Options' interest area	Query for missing — should return 0
End-to-end test	Pick 5 different degree-specialisation-interest combinations. Manually check that the 3 suggested roles make sense for a real student with that background	Human review — most important test

Part 4: The Role Change System

Students can change their target role at any time, unlimited times, with zero restrictions. Every step of the role change process is a database query — no AI calls, no cost, instant response.

4.1 The Core Principle

✓ Role changes are **UNLIMITED, FREE, and INSTANT**. The system never blocks a role change. It never says 'are you sure?' more than once. It records the change, updates the learning path, and moves on. Gaming prevention is handled by the engagement score (visible only to placement officers) and the Skills Passport eligibility threshold — **NOT** by restricting role changes.

4.2 What Happens When a Student Changes Roles

Here is the exact sequence of operations, in order:

Step 1: Student Browses Roles

The student clicks 'Explore Other Roles.' The platform loads the role catalogue — all 250 roles, filterable by job family, AI exposure level, and zone (Green/Amber/Red for their degree). This is one database query joining roles with degree_role_zones.

```
SELECT r.*, drz.zone
FROM roles r
LEFT JOIN degree_role_zones drz
  ON r.role_id = drz.role_id
  AND drz.degree_type = [student degree]
  AND drz.specialisation = [student specialisation]
WHERE r.is_active = TRUE
ORDER BY drz.zone, r.job_family, r.role_name
```

The student can filter by job family, zone, or AI exposure level. These are client-side filters on the already-loaded data — no additional queries needed.

Step 2: Student Clicks on a Role

The platform shows the role profile plus a comparison to their current situation:

```
-- Get skills for the new role
SELECT s.skill_name, rs.importance
FROM role_skills rs JOIN skills s ON rs.skill_id = s.skill_id
WHERE rs.role_id = [new role id]

-- Get student completed skills
SELECT skill_id FROM student_skill_completions
WHERE student_id = [student id]

-- Compare in application code:
-- skills_transfer = intersection of completed skills and
--                      new role skills
-- coverage = skills_transfer count / new role total skills
-- to_develop = new role skills minus completed skills
```

The student sees: 'You have 4 of 13 skills for this role. 6 skills transfer from your current role. Zone: Green (strong match for your degree).' This is 2 database queries plus arithmetic in the application code.

Step 3: Student Clicks 'Switch to This Role'

The confirmation screen shows all the information calculated in Step 2 plus:

Your quotient scores carry over: CQ 58, EQ 62, AQ 45, XQ 51, LQ 40, AIQ 35 (these never change when you switch roles — they measure YOU, not the role).

What transfers: 6 of your completed skills are also needed for this role. You start with 46% coverage.

What does not transfer: 7 new skills to develop. Estimated time: 8-12 weeks at 3 hours per week.

Your previous role: Your progress on Financial Analyst (62% coverage, 45 days) will be archived. You can return to it anytime and your progress will be recalculated based on your current skills.

Step 4: Student Confirms the Switch

The system executes these database operations:

```
-- 1. Archive the old role
INSERT INTO role_archive (student_id, role_id,
    skills_covered_at_archive, total_skills_at_archive,
    days_on_role, archived_date)
VALUES ([student], [old role], [covered count],
    [total count], DATEDIFF(NOW(), primary_role_start_date),
    NOW())

-- 2. Update the student record
UPDATE students SET
    primary_role_id = [new role id],
    primary_role_start_date = NOW(),
    total_role_changes = total_role_changes + 1
WHERE student_id = [student]
```

That is it. Two database writes. Under 50 milliseconds. No AI. No cost.

Step 5: Dashboard Updates Automatically

The learning path recalculates using the priority formula:

```
For each uncompleted skill in the new role:
    priority = 0

    -- Role Importance Points
    IF skill importance = "High": priority += 10
    IF skill importance = "Medium": priority += 7
    IF skill importance = "Low": priority += 4

    -- Cross-Role Bonus (if student has secondary role)
    IF skill also needed for secondary role: priority += 5
    IF skill needed for both secondary AND alternative:
        priority += 10
```

```
-- Almost-There Bonus
IF student completed a prerequisite for this skill:
    priority += 8

Sort skills by priority (highest first)
This is the new learning path
```

This calculation runs entirely in the application code using data from 3 database queries (new role skills, student completions, secondary role skills). No AI calls. Under 200 milliseconds.

4.3 Returning to an Archived Role

If a student wants to return to a role they previously had, they go to 'My Archived Roles', see their list of previous roles with the coverage they had when they left, and click 'Reactivate.' The system does NOT restore the old coverage. It RECALCULATES using the student's CURRENT skills. This means if the student learned new skills while on other roles that are also relevant to the old role, their coverage for the old role may be HIGHER than when they left it.

```
-- Reactivation query
-- Same as switching to any role — archive current,
-- set the reactivated role as primary
-- Coverage is recalculated fresh from current skills
```

4.4 Performance at Scale

Here is the stress test calculation:

Scenario	Database Operations	Time
1 student changes role	2 writes + 3 reads = 5 operations	Under 500ms
100 students change simultaneously	500 operations	Under 2 seconds
50,000 students ALL change role on same day	250,000 operations spread across hours	Database handles 3,000-5,000 ops/second — completes easily
1 student changes role 50 times in one day	250 operations over hours	Invisible to database. Engagement score flags the behaviour to placement officer

A standard PostgreSQL database on a modest cloud server (AWS RDS db.t3.medium, approximately \$50-70 per month) handles all of this without any special architecture, caching, or optimisation.

4.5 Gaming Prevention (No Restrictions Needed)

Four mechanisms prevent gaming without restricting student freedom:

Mechanism 1 — Engagement Score: The Role Stability component of the engagement formula (weighted 15%) decreases with frequent role changes. The placement officer sees this as Green/Amber/Red. The student never sees the raw engagement score.

Mechanism 2 — Skills Passport Threshold: To generate a Skills Passport, the student needs 30% skill coverage + 3 completed assessments + 30 days on the current role + English passed. A student who changes roles every week never hits the 30-day requirement.

Mechanism 3 — Visible Progress Statement: Before every switch, the student sees: 'You have spent 12 days on Financial Analyst and covered 35% of skills. Switching now will archive this progress.' This is informational, not a restriction.

Mechanism 4 — Soft Cooling Recommendation: If a student has changed roles more than 3 times in 30 days, the system shows: 'You have explored several roles recently. Consider spending more time on your current role to build depth. Most students who achieve strong placement preparation spend at least 4 weeks on a role.' This is a suggestion, not a block.

4.6 Role Change Exceptions

Exception	What Happens	Solution
Student switches to a role and that role is later deactivated (removed from platform in quarterly update)	The student's <code>primary_role_id</code> points to an inactive role	When a role is deactivated, notify affected students: 'Your target role [X] has been updated. We recommend [Y] as a similar alternative.' Automatically suggest the closest active role but let the student choose.
Student has completed all skills for a role, then quarterly update adds new skills to that role	Student's coverage drops from 100% to (say) $14/15 = 93\%$	Show notification: '1 new skill has been added to your role profile. You are now at 14 of 15 skills.' This is normal platform operation, not an error.
Student's degree-specialisation is not in the matching table (new degree programme at a new college)	The matching engine returns 0 results	Fallback: show the 'General / Exploring Options' recommendations for the closest degree type. Log the missing degree-specialisation so you can add it in the next update. Never show an empty recommendation screen.
Student switches to a Red zone role (stretch for their degree)	The switch proceeds normally	Show an advisory: 'This role is a stretch for your degree background. You may need additional preparation. Your placement officer can advise.' The switch is NOT blocked. Red zone is information, not a restriction.
Student's archived role data grows very large (they changed roles 100+ times)	The <code>role_archive</code> table has 100+ rows for this student	No problem. Each row is approximately 100 bytes. 100 rows = 10KB. The database does not notice. The student's archive screen may need pagination if it shows more than 20 archived roles.

4.7 Testing the Role Change System

Test	Steps	Expected Result
Basic role switch	Create test student with role A. Switch to role B. Check database.	students table: <code>primary_role_id</code> = B, <code>total_role_changes</code> = 1. <code>role_archive</code> : 1 row for role A with correct coverage and days.

Skill transfer	Create student with 5 completed skills. 3 of these are shared between role A and role B. Switch from A to B.	Coverage for role B correctly shows the 3 transferred skills. Learning path shows only the skills NOT already completed.
Multiple switches	Switch the same student through roles A → B → C → A (back to original).	role_archive has 3 rows (A, B, C). When returning to A, coverage is recalculated from current skills (may be higher than original if student learned new skills while on B and C).
Engagement score impact	Create student. Switch roles 5 times in rapid succession.	Engagement score Role Stability component should be low. Overall engagement indicator should be Amber or Red.
Passport eligibility reset	Student has 35% coverage and 40 days on role A (passport eligible). Switch to role B.	Passport eligibility resets: days_on_role for B starts at 0. Student must spend 30 days on B before becoming passport-eligible again, even if coverage is immediately above 30%.
Red zone advisory	Student with B.Com Finance degree switches to Software Developer (Red zone).	Advisory message appears. Switch is not blocked. Zone shows Red on the student's dashboard.
Deactivated role handling	Admin deactivates a role that 10 test students are targeting.	All 10 students receive notification. Their dashboard shows the deactivation message with a suggested alternative.

Part 5: The Learning Path Engine

The learning path shows each student which skills to develop next, in priority order, with recommended courses for each. It recalculates automatically whenever the student completes a skill, changes roles, or when the platform's quarterly update adds new skills.

5.1 How the Learning Path is Calculated

The learning path is NOT stored in the database. It is calculated on-the-fly every time the student views their dashboard. This means it is always current — if the student completed a skill 5 minutes ago, the learning path immediately reflects that.

The Calculation (In Plain English)

Step 1: Get all skills for the student's primary role from `role_skills` table.

Step 2: Get all skills the student has completed from `student_skill_completions` table.

Step 3: Calculate the difference: `uncompleted_skills` = role skills minus completed skills.

Step 4: For each uncompleted skill, calculate a priority score using this formula:

```
priority = 0

Role Importance (from role_skills.importance):
  High = 10 points
  Medium = 7 points
  Low = 4 points

Cross-Role Bonus (if student has secondary role):
  Skill needed for 2 roles = +5 points
  Skill needed for 3 roles = +10 points

Almost-There Bonus:
  Student completed a prerequisite = +8 points
  (A prerequisite is a skill in the same skill_category
   that is marked as a dependency in the skills table)
```

Step 5: Sort uncompleted skills by priority (highest first). This sorted list IS the learning path.

Step 6: For each skill in the learning path, look up the recommended course from the `courses` table. Show the course name and link.

5.2 What the Student Sees

The learning path on the dashboard shows a prioritised list. Each item shows: the skill name, why it matters (e.g., 'High importance for Financial Analyst'), the recommended course with a direct link, estimated time to complete, and a 'Mark as Complete' button (for self-reported completions) or automatic completion tracking (for platform assessments).

5.3 When the Learning Path Changes

Event	What Changes	How
Student completes a skill	That skill is removed from the learning path. Next highest-priority skill moves to top.	Automatic — the calculation excludes completed skills
Student changes primary role	Entire learning path recalculates for the new role. Skills completed for the old role that are shared with the new role still count.	Automatic — triggers recalculation on next dashboard load
Student adds a secondary role	Cross-Role Bonus activates. Skills shared between primary and secondary roles get +5 priority points and move up in the list.	Automatic — recalculation picks up secondary role
Quarterly update adds new skill to a role	The new skill appears in the learning path at its calculated priority position. Student's coverage percentage may decrease.	Automatic — new skill is in role_skills, so it appears in the next calculation
A recommended course becomes unavailable	The backup course (if one exists) is shown instead. If no backup, a message says 'We are finding a new course for this skill.'	Automatic — course availability check is part of the query

5.4 Performance

The learning path calculation requires 3 database queries (role skills, student completions, courses) plus arithmetic in the application code. Total time: under 200 milliseconds. This runs on every dashboard load, but since the data changes rarely (only when the student completes a skill or changes roles), the result can be cached on the client for the duration of the session and refreshed only when a change occurs.

5.5 Testing the Learning Path

Test	Steps	Expected Result
New student, no completions	Create student targeting Financial Analyst with 0 completed skills	Learning path shows all 13-15 skills, sorted by priority. Highest-importance skills are at the top.
Partial completion	Mark 5 skills as completed for the same student	Learning path shows only the remaining 8-10 skills. The 5 completed skills are not in the list.
Priority ordering	Check that a High importance skill (10 pts) appears above a Medium skill (7 pts) when no other bonuses apply	High skills at top, Medium in middle, Low at bottom
Cross-role bonus	Student has primary (Financial Analyst) and secondary (Business Analyst). A skill shared by both roles should rank higher.	The shared skill gets +5 bonus and appears higher than a non-shared skill of the same importance level
Complete all skills	Mark all skills as completed for a role	Learning path shows: 'Congratulations! You have covered all skills for Financial Analyst.' with a prompt to generate their

		Skills Passport
Course link works	Click the recommended course link for each skill in the top 5	Each link opens the correct course page (Coursera, Google, NPTEL, or Microsoft Learn)

Part 6: Skills Passport Generation

The Skills Passport is a verified capability document that employers use to assess candidates. It is the key output of the entire platform. This section covers when it is generated, what it contains, and how the generation works.

6.1 Eligibility Threshold

A student can generate a Skills Passport ONLY when ALL four conditions are met:

Condition 1: 30% or higher skill coverage for their primary role. If the role has 15 skills, the student must have completed at least 5.

Condition 2: At least 3 quotient assessments completed (any 3 of the 7 quotients, at any tier).

Condition 3: At least 30 days on the current role. This prevents students from switching to an easy role, generating a passport, and switching back.

Condition 4: English for Work diagnostic passed (Band A or Band B completion).

The eligibility check is 4 database queries — one per condition. No AI. Instant.

```
-- Condition 1: Skill coverage
completed = COUNT(*) FROM student_skill_completions WHERE
  student_id = X AND skill_id IN (role_skills)
total = COUNT(*) FROM role_skills WHERE role_id = Y
coverage = completed / total >= 0.30

-- Condition 2: Assessment count
assessments = COUNT(*) FROM student_quotient_scores WHERE
  student_id = X AND score IS NOT NULL
has_enough = assessments >= 3

-- Condition 3: Days on role
days = DATEDIFF(NOW(), primary_role_start_date)
long_enough = days >= 30

-- Condition 4: English passed
english = SELECT english_status FROM students WHERE id = X
passed = english IN ("Band A", "Band B Complete")

-- Eligible if ALL four are true
```

6.2 What the Skills Passport Contains

The passport is a structured document (displayed on screen and downloadable as PDF) with these sections:

Header: Student name, college, degree, target role, date generated.

Capability Radar Chart: The 7-quotient radar chart showing the student's scores. Generated as an SVG or image from the quotient scores — no AI needed.

Quotient Scores Table: Each quotient with its score and descriptor (Foundation/Developing/Competent/Strong/Exceptional). Pure data display.

Skill Coverage: 'X of Y skills developed for [role name].' Lists covered skills and skills in progress. Pure data display.

AI Collaboration Evidence: If the student completed the AIQ module, their AI Fluency sub-scores: Iteration Fluency, Critical Evaluation, Collaboration Design. Pure data display.

Narrative Summary: A 3-4 sentence paragraph summarising the student's strengths and readiness. THIS IS THE ONLY PART THAT USES CLAUDE.

Mandatory Disclosure: 'This Skills Passport is an evidence-based capability profile reflecting the candidate's verified preparation. This is not a certification of employment readiness or a guarantee of job placement.' This text is HARDCODED, never generated by AI.

6.3 Generating the Narrative (The Only AI Call)

The narrative summary is the only part that calls the Claude API. Here is the exact prompt:

```
Write a 3-4 sentence professional summary for a Skills Passport.
This summary will be read by employers evaluating the candidate.
```

```
Student: [name]
Degree: [B.Com Finance]
Target Role: [Financial Analyst]
Quotient Scores: CQ [65], EQ [58], AQ [52], XQ [61],
  LQ [45], AIQ [55], DQ [72]
Skills Covered: [10] of [14] for Financial Analyst
Strongest Quotient: [DQ at 72 — Domain Knowledge]
Area for Growth: [LQ at 45 — Leadership]
AI Fluency Score: [55]
```

```
Write in third person. Be factual. Do not exaggerate.
Do not use words like "exceptional" or "outstanding" unless
the score is above 85. Never guarantee employment outcomes.
Maximum 4 sentences, maximum 80 words total.
```

This call uses approximately 300 input tokens and 100 output tokens. Cost: approximately \$0.002 (0.2 cents) per passport. At 15,000 passports per semester, the total cost is approximately \$30.

6.4 Caching the Passport

✓ Once generated, the Skills Passport narrative is CACHED in the database. It is only regenerated when the student's data changes significantly (a quotient score changes by more than 5 points, or skill coverage changes by more than 10%). This means most students generate one API call total, not one per view.

```
-- Store in generated_profiles table
```

```

INSERT INTO generated_profiles (student_id, role_id,
    narrative_text, generated_date,
    scores_snapshot, coverage_snapshot)
VALUES ([student], [role], [Claude output], NOW(),
    [JSON of current scores], [current coverage %])

-- On next view, check if recalculation needed:
-- IF current scores differ from snapshot by > 5 points
-- OR current coverage differs by > 10%
-- THEN regenerate. ELSE show cached version.

```

6.5 Testing the Skills Passport

Test	Steps	Expected Result
Eligibility check — all pass	Create student with 35% coverage, 4 assessments, 40 days, English passed	Generate Passport button is enabled and active
Eligibility check — coverage too low	Create student with 20% coverage but other conditions met	Button disabled. Message: 'Complete more skills to reach 30% coverage (currently 20%)'
Eligibility check — not enough days	Create student with 50% coverage, 4 assessments, 15 days, English passed	Button disabled. Message: 'Spend at least 30 days on your current role (currently 15 days)'
Narrative generation	Generate passport for eligible student. Read the narrative.	Narrative is factual, mentions strengths and growth areas correctly, does not exaggerate, under 80 words
Caching works	Generate passport, then view it again without changes	Second view loads instantly (no API call). Check server logs to confirm no Claude API call was made.
Cache invalidation	Generate passport. Then complete 2 more skills. View passport again.	If coverage changed by more than 10%, a new narrative is generated. If less than 10%, cached version shown.
Mandatory disclosure	Check every generated passport	The legal disclosure text appears at the bottom of every single passport, unchanged, hardcoded
Score cap	Create student with quotient score of 98	Passport shows maximum of 95. No student ever sees 100%.

Part 7: Engagement Score Calculation

The engagement score measures how actively each student uses the platform. It is visible **ONLY** to placement officers, never to students. It appears as Green (70+), Amber (40-69), or Red (below 40) on the placement officer dashboard.

7.1 The Formula

```
Engagement Score =
  (Learning Progress x 0.30)
+ (Assessment Completion x 0.25)
+ (Activity Recency x 0.20)
+ (Role Stability x 0.15)
+ (Course Completion Rate x 0.10)

Learning Progress = (skills completed / total skills) x 100

Assessment Completion = (assessments done / assessments
                        available) x 100

Activity Recency:
  Active in last 7 days = 100
  Active in last 14 days = 75
  Active in last 30 days = 50
  Active in last 60 days = 25
  Not active for 60+ days = 0

Role Stability:
  0-1 role changes in last 60 days = 100
  2-3 role changes = 70
  4-5 role changes = 40
  6+ role changes = 10

Course Completion Rate = (courses completed / courses
                        started) x 100
```

7.2 When to Calculate

Calculate the engagement score weekly (every Sunday night) as a batch job for all active students. Store the latest score in a student_engagement table with student_id, score, calculated_date, and indicator (Green/Amber/Red). The placement officer dashboard reads from this table.

Do NOT calculate in real-time on every page load — the batch calculation is simpler and sufficient for the placement officer's needs.

7.3 Testing

Test	Student Profile	Expected Score Range
Highly engaged	60% coverage, 5 assessments, active today,	85-95 (Green)

student	0 role changes, 80% course completion	
Moderate student	30% coverage, 2 assessments, active 2 weeks ago, 1 role change, 50% course completion	45-60 (Amber)
Disengaged student	10% coverage, 0 assessments, not active for 45 days, 0 role changes, 0% completion	10-20 (Red)
Role switcher	40% coverage, 3 assessments, active today, 6 role changes, 60% completion	55-65 (Amber — dragged down by Role Stability)

Part 8: Every Exception and Edge Case

This section lists every unusual situation that could arise and the exact solution for each. If it is not listed here, it is not an exception — handle it with the standard logic described in Parts 1-7.

Exception	When It Happens	Solution
Student's degree not in matching table	New college joined with a degree programme you have not mapped yet	Show the 5 most popular roles across ALL degrees as a generic recommendation. Log the missing degree-specialisation. Add it to the matching table within 48 hours.
Student selects a role with zero courses mapped	A role has skills that do not have recommended courses yet	Show the skill in the learning path with message: 'Search for [skill name] on Coursera or Google to find free courses. We are adding a recommended course soon.' Never hide a skill because it lacks a course.
Student completes a skill that was removed from role in quarterly update	Quarterly update removed a skill from a role because it is no longer relevant	The completion record stays in student_skill_completions permanently. The skill simply no longer appears in the learning path because it is no longer in role_skills. The student's coverage percentage may increase because the total skills decreased.
Two students at the same college get different recommendations for the same degree + interest	This should NOT happen — the matching table is deterministic	If this happens, it is a bug. Debug: check that both students have exactly the same degree_type, specialisation, and interest_area values (including capitalization and spaces). A trailing space in a text field can cause a mismatch.
Claude API is down when student tries to generate Skills Passport	Rare but possible — API outage	Show message: 'Your Skills Passport is being prepared. Please check back in a few minutes.' Queue the generation request. Retry every 5 minutes for up to 1 hour. If still failing, generate a passport WITHOUT the narrative paragraph (all other sections are data-only and do not need AI). Add the narrative when the API recovers.
Student has dual specialisation (e.g., B.Com Honours in Finance and Marketing)	Some colleges offer dual specialisations	Use the student's PRIMARY specialisation (the one they select first during onboarding) for matching. Show an additional set of recommendations for the secondary specialisation in a 'Based on your second specialisation' section. Both sets come from the matching table — just run two queries instead of one.
College adds a new degree programme mid-year	Partner college introduces a new programme after the platform launched	Add the degree-specialisation to degree_interest_areas and degree_role_recommendations within 1 week. Use Claude to generate the matching data (same prompt as initial generation). Deploy as a database update — no code change needed.
Student wants a role not in the 250-role catalogue	Student searches for 'Ethical Hacker' but this specific role does not exist	Show: 'This role is not yet in our catalogue. Here are similar roles: [Cybersecurity Analyst, Security Operations Analyst]. We have noted your interest and may add this role in the future.' Log the request in a role_requests table. If 10+ students request the same role, generate a profile for it.
A course URL breaks between quarterly checks	Detected by student reporting a dead link	Immediately mark the course as is_active = FALSE. If backup_course_id exists, the system automatically shows the backup. If no backup, show the 'search for this skill on Coursera' message. Fix within 48 hours.

		This is the ONLY type of fix that happens outside the quarterly cycle.
Student's quotient score decreases after retaking an assessment	Student retakes an assessment and scores lower than before	Keep the HIGHER score. The student's profile shows their best performance. The lower score is logged for internal analytics but never displayed. This prevents anxiety about retaking assessments.
Placement officer wants to see a student's full role change history	Officer is evaluating a student's engagement pattern	Show the role_archive table for that student: list of all previous roles with dates, days spent, and coverage at time of archive. Show total_role_changes count. Show the engagement score with the Role Stability component highlighted.
Platform needs to support a new language (Hindi, Tamil, etc.)	College partners in regions where English is not the primary language	Phase 1: Keep the platform in English (it is an English-for-work platform). Phase 2 (Year 2+): Add UI translations for navigation and instructions only. Role data, skill names, and narratives remain in English because employers expect English.

Part 9: The Exact Build Sequence

Here is the order in which to build everything, day by day.

Week 1: Role Data Generation

Day	Task	Output
Day 1	Set up Claude API account. Add \$10 credits. Prepare roles.csv with first 100 role names (Batch 1). Run Step 1 prompt for all 100 roles.	100 JSON files with structured role data
Day 2	Run Step 2 prompt for all 100 roles. Begin verification of AI tool names and salary ranges for Batch 1.	100 narrative texts. Verification started.
Day 3	Complete verification for Batch 1. Load verified data into database. Run the test queries from Section 2.9.	100 verified role profiles in the database, all tests passing
Day 4	Generate Batch 2 (roles 101-175). Begin verification.	75 more role profiles generated
Day 5	Generate Batch 3 (roles 176-250). Complete verification for Batch 2.	All 250 roles generated. 175 verified.

Week 2: Matching Engine and Remaining Data

Day	Task	Output
Day 6	Complete verification for Batch 3. All 250 roles verified and in database.	250 verified role profiles. All test queries passing.
Day 7	Generate the degree_interest_areas table (all interest area dropdowns for all degree-specialisation combinations). Generate degree_role_recommendations table (matching data).	400-600 interest area entries. 1,200-1,800 matching entries.
Day 8	Generate degree_role_zones table (Green/Amber/Red for every degree-role combination). Verify matching data with human spot-checks.	Zone classifications for all 250 roles x all degree types
Day 9	Build the matching engine runtime code: onboarding flow, interest area dropdown, recommendation display, role selection. Run all matching tests from Section 3.7.	Working onboarding flow that suggests 3 roles
Day 10	Build the role catalogue browser: filterable list of all 250 roles with zone indicators, AI exposure levels, and search.	Working role catalogue that students can browse

Week 3: Role Changes, Learning Path, and Passport

Day	Task	Output
Day 11	Build the role change system: browse, compare, confirm, switch, archive. Run all role change tests from Section 4.7.	Working role change flow
Day 12	Build the learning path engine: priority calculation, skill display, course recommendations. Run all learning path tests from Section 5.5.	Working learning path on student dashboard
Day 13	Build the Skills Passport: eligibility check, data display, Claude API integration for narrative, caching. Run all	Working Skills Passport generation

	passport tests from Section 6.5.	
Day 14	Build the engagement score calculator: batch job, formula implementation, placement officer dashboard display. Run all engagement tests from Section 7.3.	Working engagement scores visible to placement officers
Day 15	End-to-end integration testing. Create 10 test students with different profiles. Walk each through the full journey: onboard → select role → use learning path → complete skills → change role → generate passport.	10 complete student journeys tested successfully

Week 4: Pilot and Scale

Day	Task	Output
Day 16-17	Fix bugs found in integration testing. Performance testing: simulate 1,000 concurrent users using a load testing tool. Verify response times are under 500ms for all operations.	All bugs fixed. Performance confirmed.
Day 18-19	Open platform to first 2-3 colleges (pilot). Monitor for issues. Watch server logs. Talk to placement officers and students for feedback.	Pilot running with real students
Day 20	Fix any issues found during pilot. Prepare for full launch.	Pilot confirmed stable

After Week 4, open the platform to all 25 colleges. You have a working system serving 50,000 students with 250 verified role profiles, intelligent matching, unlimited role changes, personalised learning paths, and verified Skills Passports.

Part 10: Complete Cost Summary

10.1 Build Phase (One-Time)

Item	Cost
Claude API for role data generation (250 roles)	\$9 (₹750)
Claude API for matching table generation	\$5 (₹420)
Claude API for zone classification generation	\$3 (₹250)
Claude API for testing and iteration (3x buffer)	\$15 (₹1,250)
Total Claude API cost for complete data generation	\$32 (₹2,670)

10.2 Runtime Phase (Ongoing, Per Semester)

Item	Volume	Cost Per Semester
Role browsing, matching, role changes, learning paths	Unlimited for 50,000 students	₹0 (all database queries)
Quotient assessment scoring (Tier 2)	20,000 assessments at \$0.005 each	\$100 (₹8,400)
Skills Passport narrative generation	15,000 passports at \$0.002 each	\$30 (₹2,500)
Total runtime AI cost per semester	50,000 students	\$130 (₹10,900)
Total runtime AI cost per year	50,000 students, 2 semesters	\$260 (₹21,800)

10.3 Infrastructure (Monthly)

Item	Specification	Monthly Cost
Database server	PostgreSQL on AWS RDS db.t3.medium (or equivalent)	₹4,000-5,000
Application server	2 vCPU, 4GB RAM (AWS EC2 t3.medium or equivalent)	₹3,000-4,000
Storage	50GB for database + 10GB for files	₹500-1,000
CDN and static hosting	For frontend assets	₹500-1,000
Total monthly infrastructure		₹8,000-11,000
Total annual infrastructure		₹96,000-1,32,000

10.4 Total Annual Operating Cost

Item	Annual Cost
AI API costs (Claude)	₹21,800
Infrastructure	₹96,000-1,32,000

Domain, SSL, email	₹5,000
Total annual operating cost for 50,000 students	₹1,23,000-1,59,000 (approximately ₹1.2-1.6 lakh)

Revenue from 25 colleges at ₹2.5 lakh average = ₹62.5 lakh per year. Operating cost = ₹1.2-1.6 lakh per year. Operating margin: over 97%. This is a commercially viable and highly profitable business.

Part 11: Pre-Launch Checklist

Before going live with the first college, every item on this list must be confirmed. Check each one off.

#	Item	Confirmed By	Date
1	250 role profiles loaded and verified in database		
2	All roles have 12-20 skills mapped in role_skills table		
3	AI tool names verified — no hallucinated tools in the data		
4	Salary ranges verified against AmbitionBox for all 250 roles		
5	Matching table loaded: every degree-specialisation has 5-6 interest areas		
6	Matching table verified: every interest area has exactly 3 recommendations (Primary, Secondary, Alternative)		
7	Zone classifications loaded for all degree-role combinations		
8	Onboarding flow tested: student can select degree, specialisation, interest → receives 3 role suggestions		
9	Role catalogue browsable: all 250 roles visible with filters for job family, zone, AI exposure		
10	Role change tested: student can switch roles, archive works, learning path recalculates		
11	Learning path shows prioritised skills with course links for the top 10 most popular roles		
12	Skills Passport generates correctly for eligible students with cached narrative		
13	Engagement score calculates correctly for 5 test profiles covering Green, Amber, and Red		
14	Claude API fallback works: if API is down, passport generates without narrative and queues for retry		
15	Missing degree fallback works: students with unmapped degrees get generic recommendations, not an error		
16	Course link checker run: all recommended course URLs return HTTP 200		
17	Legal disclosure appears on every Skills Passport: 'This is not a certification of employment readiness...'		
18	Score cap enforced: no quotient score displays above 95 anywhere on the platform		
19	Load test passed: 1,000 concurrent users, all response times under 500ms		
20	Placement officer dashboard shows: student list, engagement scores, batch analytics, skill coverage summary		

When every row in this checklist has a name and a date, you are ready to go live. Not before.